Env-AI-Ron

Ecological planting blueprint: Lagos Mainland (demo tile)

Generated 2026-10-11 | 100x100 cells of 20 m | objective: cooling 30%, carbon 20%, pollinator 20%, equity 30%

Demonstration data. The imagery and GIS layers are synthetic, generated to exercise the platform. They are not measurements of real Lagos.

The success measure here is not how the city looks. It is how much life, shade, clean air and carbon this land can hold, and who benefits first.

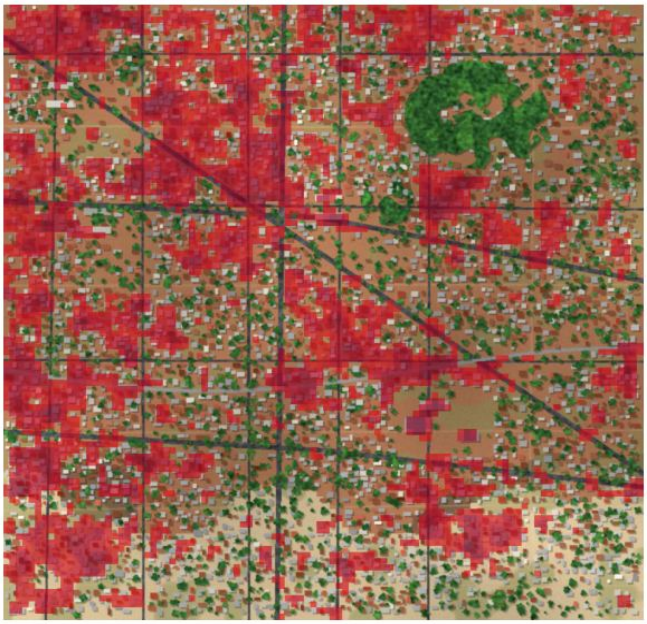
Ecological return at a glance

Native trees to plant	**404**	across	180 sites / 7.33 ha
Air cooling at planted sites	**0.31 C**	by year 20	0.06 (y5), 0.23 (y10)
Cooling, 100 m neighbourhood	**0.28 C**	by year 20	peak 0.91 C
Carbon stored	**115 tCO2**	by year 20	28 (y10)
New canopy	**4.2 ha**	by year 20	cover 13.07% to 14.11%
Habitat quality index	**48.4**	by year 20	from 43.8 today
Pollinator groups supported	**11**	of 12 tracked	from the species palette
Health equity	**72%**	of trees	in high-vulnerability areas (only 26% of plantable land is)
Species diversity	**7 species**	Shannon 1.63	no species above 35%

Reading the land: what is there now

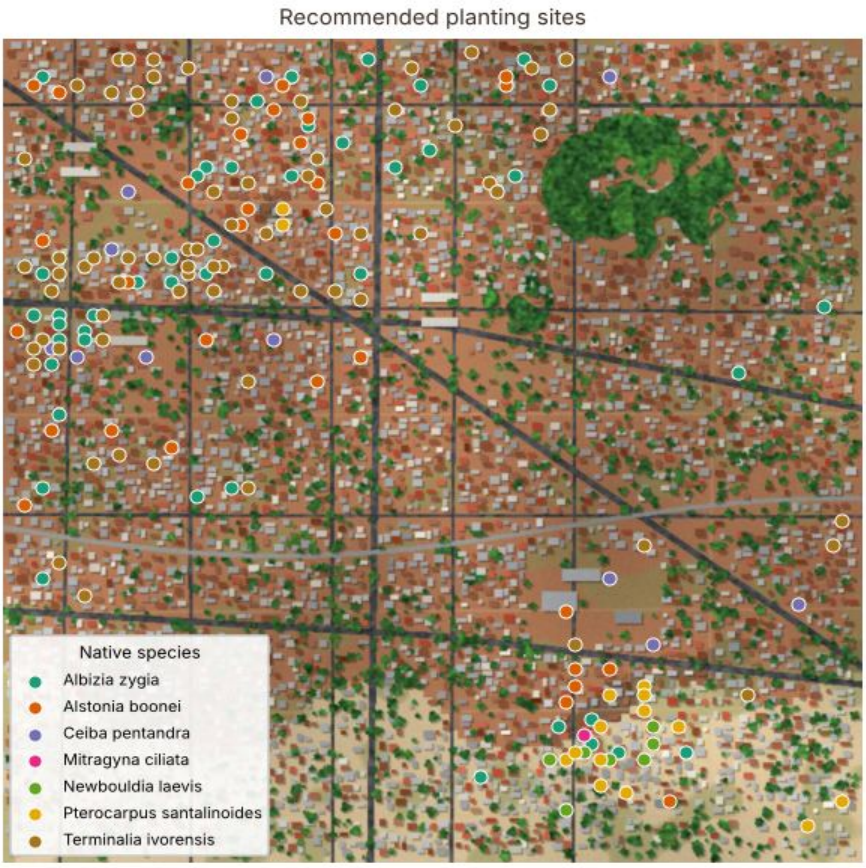
Over 407.1 ha, the imagery reads as 28.2% concrete and other hard surface, 13.1% tree canopy and 58.8% bare soil. Roughly 122.1 ha qualify as **ecological voids**: the hottest cells (a modelled surface-heat proxy, mean 34.3 C, hottest decile 35.6 C) where canopy is thin and concrete dominates. About 264.2 ha of bare ground is plantable without removing any existing tree, building, road or drainage channel.

Ecological voids: hot, canopy-poor, concrete-dominated ground



The blueprint: where to plant

Sites were chosen by a genetic algorithm that balances cooling, carbon, pollinator connectivity and health equity. Cooling saturates (a hundred trees in one place help less than the same trees spread across hot ground) while pollinators need stepping stones, so the objectives pull against each other, and the search finds the compromise.



Composite score: genetic algorithm 1.015 vs greedy top-N 0.398 (0.918 before the fairness penalty) vs random 0.819 (1.0 = matches a plan that chases each goal alone; a fairness penalty applies if one ward would receive over 30% of sites).

Ecological ROI by planting cluster

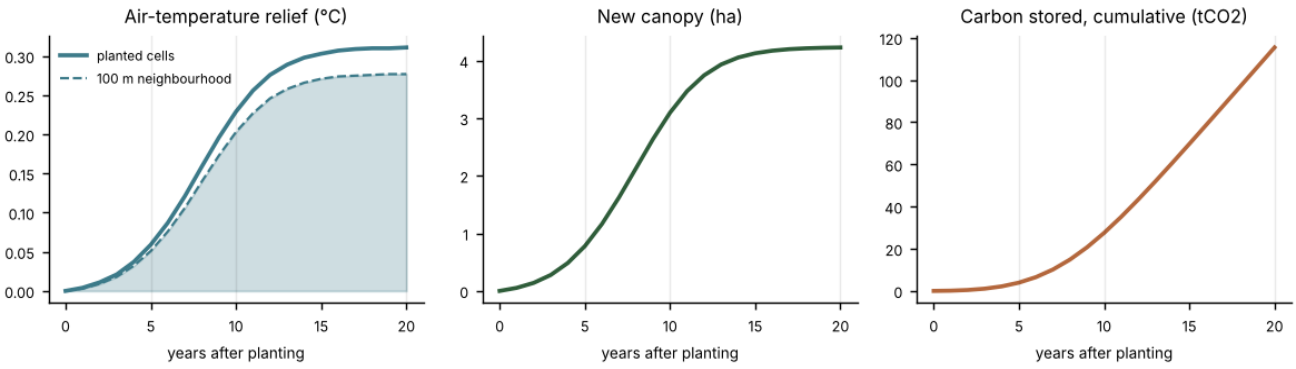
Cluster	Trees	Dominant native species	Cooling y20	tCO2 y20	Pollinator groups
C133	17	Newbouldia laevis	0.56 C	3.7	9
C135	12	Newbouldia laevis	0.49 C	2.4	7
C86	9	Terminalia ivorensis	0.69 C	2.9	9
C134	7	Newbouldia laevis	0.21 C	1.0	4
C131	7	Newbouldia laevis	0.21 C	1.0	4
C138	7	Newbouldia laevis	0.21 C	1.0	4
C38	6	Albizia zygia	0.49 C	1.9	5
C68	6	Terminalia ivorensis	0.53 C	1.9	6

C133. Planting 17 native trees here (mostly Akoko / Ogirisako, Newbouldia laevis) is projected to cool the planted cells by about 0.6 °C by year 20 (range 0.3-0.8 °C), store ~3.7 tCO2 and support 9 pollinator groups.

C135. Planting 12 native trees here (mostly Akoko / Ogirisako, Newbouldia laevis) is projected to cool the planted cells by about 0.5 °C by year 20 (range 0.3-0.7 °C), store ~2.4 tCO2 and support 7 pollinator groups.

C86. Planting 9 native trees here (mostly Idigbo, Terminalia ivorensis) is projected to cool the planted cells by about 0.7 °C by year 20 (range 0.4-1.0 °C), store ~2.9 tCO2 and support 9 pollinator groups.

Projected recovery over 20 years



Shaded range on the cooling chart reflects the stated uncertainty of the cooling coefficient; see assumptions.

Species chosen, and species refused

Every recommended tree is native, matched to the cell's soil and flood risk, and the palette is split so that no single species dominates (a monoculture is one disease away from losing its canopy).

Species	Trees	Common name
Terminalia ivorensis	*140*	*Idigbo*
Albizia zygia	*90*	*Ayinre / Okuro*
Alstonia boonei	*60*	*Ahun / Egbu*
Newbouldia laevis	*56*	*Akoko / Ogirisako*
Pterocarpus santalinoides	*45*	*Gbengbe / Nanga*
Ceiba pentandra	*11*	*Araba / Okha*
Mitragyna ciliata	*2*	*Abura*

Refused by the engine (kept in the database so the refusal is auditable):

Species	Why it was rejected
Azadirachta indica	Invasive: displaces native flora (do no harm)
Eucalyptus camaldulensis	Invasive: displaces native flora (do no harm); High water demand: unsustainable in a water-stressed city
Prosopis juliflora	Invasive: displaces native flora (do no harm)
Delonix regia	Exotic / ornamental: low habitat value for native fauna
Roystonea regia	Exotic / ornamental: low habitat value for native fauna; High water demand: unsustainable in a water-stressed city
Casuarina equisetifolia	Invasive: displaces native flora (do no harm)

Method, assumptions and honest limits

- Cover classes come from a gradient-boosted pixel classifier (concrete / canopy / bare soil). Surface heat is a **proxy** modelled from cover fractions, not a thermal measurement.

- Cooling uses 0.35 C per +10 percentage points of canopy averaged over about 61 m, capped at 2.5 C, with a +/-40% range. These are planning assumptions to calibrate with local temperature loggers.

- Planting survival is assumed at 80%. Crown growth is a logistic curve per species.

- Species traits and pollinator associations are provisional and need review by local botanists before real planting.

- Native versus ornamental classification of *existing* greenery is not in this MVP; existing canopy is treated as habitat.

- Existing trees, buildings, roads and drainage channels are never displaced (do no harm). Street-tree planting needs a separate right-of-way workflow.

- Cells with no zoning polygon are treated as unplantable ('unzoned').

